Terra-101: How to Run Spectronaut Searches on Terra?

Convert Files to HTRMS before Your Search

Starting with snapshots v24 and v25, `panoply_spectronaut` supports converting raw files to HTRMS format before running the search.

HTRMS Converter extracts and encodes MS data from the instrument into HTRMS, a format optimized for Spectronaut, enabling faster and more cost-efficient searches. If files are converted to HTRMS format, the converted files will be saved in the `call-htrms_convert` folder within the submission output directory (i.e., the same place where the search output is stored) and can be transferred to your bucket for future searches.

New Input Variables for HTRMS Conversion

- `do_conversion` and `do_search` let you control whether to run conversion and/or Spectronaut search
- If converting files, set `files_folder` to the directory containing raw files
  - If you already have HTRMS files, set `htrms_files_folder` to that directory. When `htrms_files_folder` is defined, the workflow skips conversion and uses these converted files for the search, regardless of `do_conversion` or `files_folder`
- `convert_scheme` specifies a custom HTRMS conversion scheme exported from HTRMS Converter (BGS Factory Default is used if not defined)
- If converting files, update the `Run Label` column in the `condition_setup` file so file extensions are “.htrms”
- All other variables remain the same as in previous snapshot versions

Task name ↓	Variable	Type	
panoply_spectronaut	do_conversion	Boolean	"true" if converting to HTRMS before searching, otherwise "false"
panoply_spectronaut	do_search	Boolean	"true" if searching the data, otherwise "false"
spectronaut	experiment_name	String	
spectronaut	fasta	File	
convert_htrms	*convert_scheme*	*File*	(Optional) Path to the conversion scheme exported from HTRMS Converter
convert_htrms	*local_disk_gb*	*Int*	
convert_htrms	*num_cpus*	*Int*	
convert_htrms	*num_preemptions*	*Int*	
convert_htrms	*ram_gb*	*Int*	
panoply_spectronaut	*files_folder*	*Directory*	Directory path to the folder where raw files are stored (*If `do_conversion` is set to "false", Spectronaut will use these files for searching
panoply_spectronaut	*htrms_files_folder*	*Directory*	(Optional) Directory path to the folder where existing HTRMS files are stored (*This directory has the highest priority – if supplied, the workflow will ignore `files_folder` input and use the HTRMS directly for searching)
spectronaut	*analysis_settings*	*File*	
spectronaut	*condition_setup*	*File*	IMPORTANT! If doing HTRMS conversion before search, make sure the file extension for values in `Run Label` column is changed from ".raw/d" to ".htrms"
spectronaut	*enzyme_database*	*File*	
spectronaut	*fasta_1*	*File*	
spectronaut	*file_of_files*	*File*	
spectronaut	*json_settings*	*File*	
spectronaut	*local_disk_gb*	*Int*	
spectronaut	*num_cpus*	*Int*	
spectronaut	*num_preemptions*	*Int*	
spectronaut	*ram_gb*	*Int*	
spectronaut	*report_schema*	*File*	
spectronaut	*report_schema_1*	*File*	
spectronaut	*report_schema_2*	*File*	
spectronaut	*spectral_library*	*File*	
spectronaut	*spectral_library_1*	*File*	

gcloud CLI Cheat Sheet

To authenticate gcloud CLI, run `gcloud auth login`.

*Once ANY user has authenticated on a machine, re-authentication is not required for new command line sessions in the future.

Transferring Files, BUT FASTER!

You do NOT need to run `gcloud config set project <project-id>` before transferring files.

Once the CLI has been authenticated, files can be copied directly between different buckets with the following commands:

- Copy a file to a destination: `gcloud storage cp <source> <destination>`
- Copy multiple files to a destination: `gcloud storage cp <source-1> <source-2> ... <source-n> <destination>`
- Copy a folder to a destination: `gcloud storage cp --recursive <source> <destination>`

*The new `gcloud storage` command accelerates the transfer through parallelization.

`panoply_spectronaut` Snapshot Info

Terra does NOT track snapshot versions, so we recommend noting your snapshot version in job comment for future reference.

Snapshot	Spectronaut	Creation Date	Notes
15	v18.7	2024-06-20	
16	v19.0	2024-06-20	
17	v19.7	2024-08-15	
18	v19.0	2024-10-17	
19	v19.7	2024-11-06	Default computational parameters
20	v20.0	2025-06-16	Default computational parameters
22	v20.0	2025-09-17	Reduced computational parameters to 16 CPU cores, 64GB RAM, and 2TB disk space
23	v19.7	2025-09-17	Reduced computational parameters to 16 CPU cores, 64GB RAM, and 2TB disk space
24	v20.0	2025-09-25	Added HTRMS Conversion
25	v19.7	2025-09-25	Added HTRMS Conversion